HARDWARE APPLICATION

MODIFYING THE KEYBOARD SO CTRL AND RESET MUST BE PRESSED TO CAUSE A RESET

A notorious foible of the original Apple II was the location and shape of the RESET key. It sits next to the dash/equal key and the RETURN key where it is often accidentally pressed by the operator causing aggravation, crashes, and fear of equal signs. Additionally every one year old offspring of a thousand home enthusiasts learned where the button was that caused all the commotion. It was bad enough that various RESET lockout schemes were sold commercially, and Apple competitors advertised that you couldn't hit their RESET key accidentally. The author used to keep a cardboard cover over his RESET key and, to this day, does not reach for the upper right hand corner of the keyboard without second thoughts.

Apple pretty well solved the problem in the Apple II Plus by installing the CTRL-required slide switch on the keyboard. If this switch is in the correct position, both CTRL and RESET must be held down simultaneously to make RESET' go low. This nicely prevents accidental RESETs and takes little getting used to. The purpose of this Application Note is to show how to modify older keyboards so CTRL will be required to make the RESET key function.

As Figure 7.7 shows, both the CTRL switch and RESET switch operate by placing a ground on a line pulled up by a resistor. The modification consists of removing the ground connection from the RESET switch and connecting the RESET switch to the CTRL line. After this, the RESET switch causes the RESET' line to go low only if the CTRL line is low.

The ground is removed from the RESET switch by cutting current traces on the keyboard PC card and wire jumping the broken ground path around the RESET key. Then another wire jumper connects the CTRL line to the RESET switch. The ground connection should be made with a heavier gauge wire since ground for the entire keyboard will flow through this wire. 22 gauge wire is heavy enough. The wire jumpers can be connected to pins extending through the PC board, and they should be soldered. Keep wire length as short as possible so wires will not hang loosely. The following is an installation procedure:*

1. Separate the bottom assembly from the white case following the procedure in Appendix J.
2. Remove the keyboard from the case by removing the three screws on either end.
3. There are two ground traces which must be cut, one on each side of the PC board. Their location is shown in Figure 7.17. Cut these two traces with a pocket knife, verifying discontinuity. As shown in Figure 7.17, one trace must be peeled back slightly to form a little tail to which a wire can be connected and soldered.
4. Solder a jumper between pin 8 of the 16-pin DIP jumper socket and the peeled back ground trace. This wire should be 22 gauge.
5. Solder a jumper between the right terminal of switch 13 and pin 13 of U1 as shown in Figure 7.17.
6. The installation is complete. Operation can be verified by setting the keyboard on an insulated surface next to the base assembly, connecting the keyboard jumper to the motherboard, and applying power. After verification, disconnect the keyboard from the motherboard, install the keyboard in the white case, and install the base assembly on the white case. Don't forget to reconnect all wires, including the keyboard jumper.

*Please read the NOTE OF CAUTION at the beginning of the book before making any modification to your hardware.

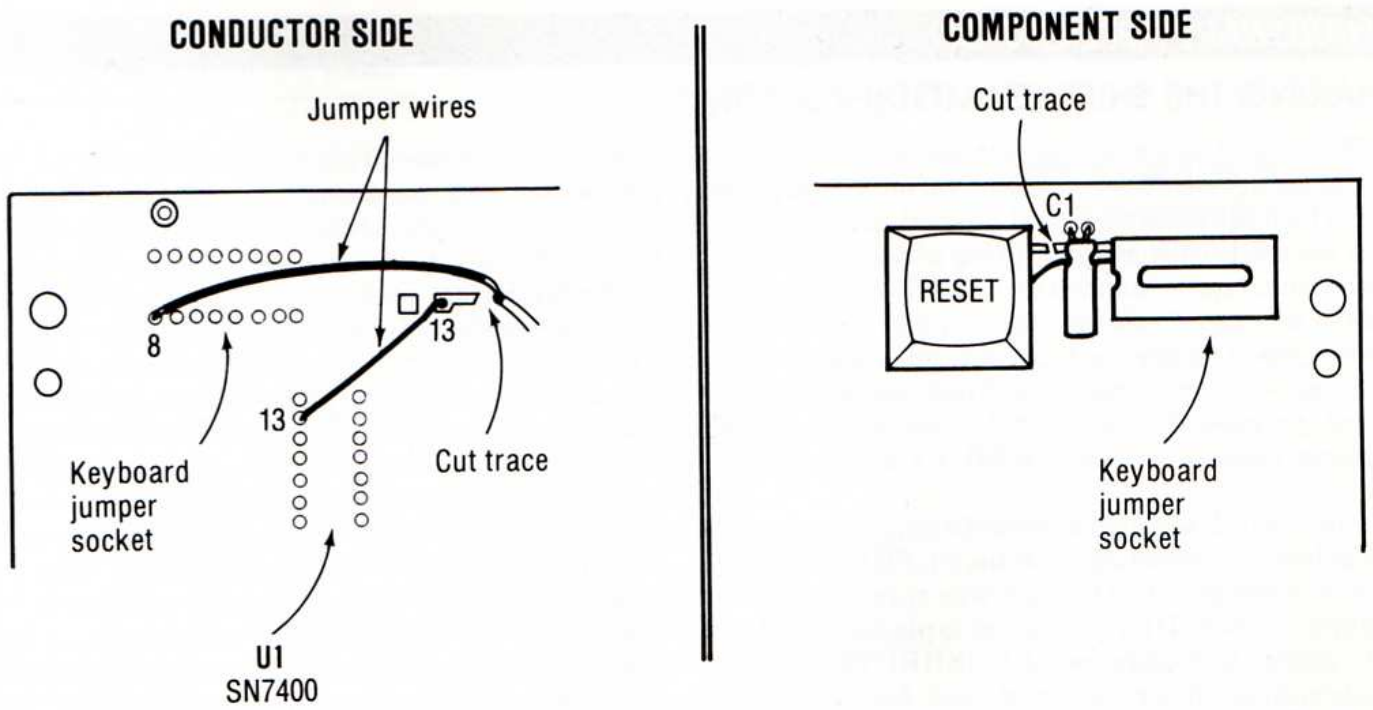


Figure 7.17 The CTRL-RESET Modification, for Older Keyboards.